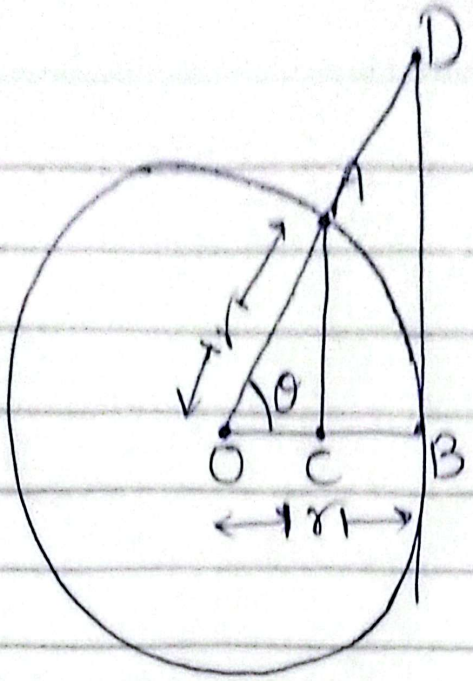


Arc AB (let AB be the arc)

∵ the circle is unit circle (whose radius as 1)

let the angle be θ which approaches to zero. i-e $\lim_{\theta \to 0} \frac{\sin\theta}{\theta} = 1$

Assume θ as very small that it approaches to zero.

Draw a ⊥ from A to the base and draw a tangent ~~from~~ passing from a point B

or

Draw two ⊥ |AC| and |DB|

Area of ΔOCA < Area of Sector ABB < Area of ΔOBD

OR

$|AC| < \text{Arc } AB < |BD|$ —— (i)

$\because \quad x = r\cos\theta$ → Base.

$\because \quad y = r\sin\theta$ → perpendicular

to find $|AC|$ → perpendicular

$\because \quad \frac{|AC|}{|OA|}$

$|AC| = r\sin\theta$

$|OA| = r$ (we know)

$|OC|$ → (We dont know)

To find Arc length AB

$\because l = r\theta$

$|AB| = r\theta$

To find $|BD|$, we dont know $|OD|$

$\because$ what we know, $|OB| = r = |OA|$

$|OB| \Rightarrow$ Base $\quad\quad$ $|BD| \Rightarrow y$

$\because \quad \tan\theta = \frac{y}{x}$ (We have to find) (We know)

$$\frac{|BD|}{|OB|} = \tan\theta$$

$$|BD| = |OB| \tan\theta$$

$$\boxed{|BD| = r \tan\theta}$$

Put values of $|BD|$, Arc AB, and $|AC|$ in eq (i)

$$r\sin\theta < r\theta < r\tan\theta$$

Divide by $r\sin\theta$ (to make it simplify)

$$1 < \frac{\theta}{\sin\theta} < \frac{1}{\cos\theta}$$

Take Reciprocal, (∵ inequality sign flips)

$$1 > \frac{\sin\theta}{\theta} > \cos\theta$$

Rewrite

$$\cos\theta < \frac{\sin\theta}{\theta} < 1$$

Take $\lim_{\theta \to 0}$

$$\cos(\theta) < \lim_{\theta \to 0} \frac{\sin\theta}{\theta} < 1 \quad \text{---- (A)}$$

$$1 < \lim_{\theta \to 0} \frac{\sin\theta}{\theta} < 1$$

Now, look we have 3 functions in (A)

$$f(x) \leq g(x) \leq h(x)$$

Now apply limit as $x \to c$

$$\lim_{x \to c} f(x) \leq \lim_{x \to c} g(x) \leq \lim_{x \to c} h(x)$$

if $\lim_{x \to c} f(x) = L$ && $\lim_{x \to c} h(x) = L$

$$L \leq \lim_{x \to c} g(x) \leq L$$

Then by squeezing

$$\lim_{x \to c} g(x) = L$$